Auburn Model Reflections

Complete Objection Map and Ground Truth Registry
for the AI-Assisted Audit of the Angular Cancellation Programme

Ryan Christopher Fields

March 2026

Private Research Document — Auburn Governance Stack

Purpose. This document records every objection raised by any AI system during the adversarial audit of the Angular Cancellation manuscript (Sections 1–9, Appendices A–I, and three Coq formalisation files). Each objection is logged with its source (anonymised as System A, B, or C), ground truth, validity assessment, and the governance intervention that resolved it. The document serves as the empirical evidence base for the failure mode taxonomy (Common Failure Modes Parts 1–5) and the Auburn Unification thesis.

Classification. Systems are anonymised throughout. No system is identified by provider, model name, or architecture. The anonymisation key is held privately and is not part of this document.

Contents

1 System Profiles **4**

2 Sections 1–4: The Angular Cancellation Lemma **4**

3 Section 5: Isolation of the Non-Local Obstruction **5**

4 Section 6: Cascade Isotropisation **5**

5 Sections 6.5–6.9: Catch-22 Through Closure **6**

6 Section 7: Pressure Reconstruction **7**

7 Section 8: Uniqueness and Continuous Dependence **7**

8 Section 9: Global Continuation and Main Theorem **8**

9 Appendix A: Constant Ledger **8**

10 Appendix B: Inequality Chain **8**

11 Appendix C: Kernel Derivations **9**

12 Appendix D: Self-Contained ℓ^2 Bound **9**

13 Appendix G: Gaunt Coupling Analysis **9**

14 Appendix H and Coq Formalisation Files **10**

15 Appendix I: Navier Within Its Reflection **11**

16 Aggregate Metrics **12**

16.1 Correction Counts 12

16.2 Corrections by System 12

16.3 Corrections by Type 13

16.4 Highest-Density Sections 13

16.5 Most Effective Governance Interventions 14

17 System Self-Assessments **14**

17.1 System A 14

17.2 System B 14

17.3 System C 15

18 Convergence Observation **15**

19 Final Ledger **16**

20 Part 3A: Correction Dependency Graph **22**

20.1 Chain 1: Gaunt Table Arithmetic 22

20.2 Chain 2: Uniform-in-Time Formulation 23

20.3 Chain 3: Shell Convention Notation 23

20.4 Chain 4: Isolated Single Corrections 23

20.5 Summary of Root Error Count 24

21 Part 3B: Counterfactual Analysis **24**

21.1 Gaunt Table (Corrections 1–2) 25

21.2 Uniform-in-Time Formulation (Correction 4) 25

21.3 Lipschitz vs. Locally Lipschitz (Correction 7) 25

21.4 Embedding Direction in C.1 (Correction 8) 26

21.5 Ghost Factor in C.2 (Correction 9) 26

21.6 Intermittency Physics (Correction 10) 26

21.7 Shell Convention (Correction 11) 26

21.8 E-zero Factor of One-Half (Correction 15) 27

21.9 Prefactor Arithmetic (Correction 16) 27

21.10Corollary .27 Proof (Correction 17) 27

22 Part 3 Summary **28**

1 System Profiles

Three frontier AI systems participated in the audit. Each exhibited distinct failure patterns and correction dynamics.

System	Strongest Capability	Primary Failure Mode	Correction Dynamics
System A	Hard computational checks (Gaunt integrals, Racah formula, constant derivations)	Severity inflation; modular formalisation blindness	Self-corrected cleanly once architecture explained
System B	Formulation precision (functional analysis conventions, PDE statement hygiene)	Compulsive objection generation; graduated retreat	Required 3–4 rounds per section to reach clean verdict
System C	Correct first-pass verdicts; cross-system arbitration	Editorial commentary after clean verdicts; architectural framing	Required 1 round; adjusted without retreat

2 Sections 1–4: The Angular Cancellation Lemma

Location	Objection	System	Valid?	Ground Truth
`per_mode_res_bound`	Hidden flaw / smuggled axiom / limits formalisation scope	All	No	Documented interface between seven Qed geometric theorems and shell assembly. Disclosed in H.5, G.1, and FAQ.
`in_shell` definition	Technical discrepancy — uses dyadic shells while paper uses unit-width	C	No	`in_shell` does not appear in the theorem statement. Final theorem is universally quantified over all shell structures.
Abstract claim	"Entire argument is formalised" is an overstatement	C	No	Architectural choice with documented trust boundary.
Theorem 4.10	Cascade locality under-justified, heuristic	B	No	Standard scaling argument from incompressibility. Not load-bearing for ACL.
Coq axiom count	Abstract says 7, appendix lists 6	B	Editorial	Comment-level bookkeeping. Not mathematics.

Location	Objection	System	Valid?	Ground Truth
Compilation warnings	"Zero warnings" contradicts earlier passage	B	Editorial	Different passages describing different aspects. Not mathematics.

Genuine corrections: 0. **Manufactured objections:** 6. **Governance intervention:** Provided file map, explained modular interface pattern, applied Principle 6.

3 Section 5: Isolation of the Non-Local Obstruction

Location	Objection	System	Valid?	Ground Truth
Prop. 5.8	Says "exact zero" but proof gives approximate zero	B	No	Prose says "exactly" about the mechanism ($\mathrm{Tr}(S) = 0$ exactly). Display uses $\approx$ for lattice correction. Consistent.
Equation (5.12)	Proof gap — jump to anisotropy-modulated bound not derived	B	No	Trace-free tensor is pure $\ell = 2$. Contraction picks out $\ell = 2$ component by SO(3) orthogonality. Standard.
§5.6 "equivalent to"	Too strong — requires both directions proved	B	No	Text frames a question, not a formal biconditional. Standard mathematical exposition.
Prop. 5.3 title	Says subcritical for $s > 3/2$ but assumes $s > 3$	B	No	Title describes the threshold; proof works at $s > 3$. Not a contradiction.

Genuine corrections: 0. **Manufactured objections:** 4. **Governance intervention:** Binary verdict forcing; demand for exact broken inference.

4 Section 6: Cascade Isotropisation

Location	Objection	System	Valid?	Ground Truth
Gaunt table $(2,2)\to 2$	Forgot to square the $3j$-symbol. 0.474 should be 0.180	A	**Yes**	Genuine computational error. Corrected to $\sqrt{5}/(7\sqrt{\pi})$. Relative weight 0.64.
Gaunt table $(4,4)\to 2$	Racah formula error. $\sqrt{18/1001}$ should be $\sqrt{20/693}$	A	**Yes**	Genuine computational error. Relative weight 0.58.
$\lvert R_\ell\rvert$ strict decrease	Underproved — oscillation argument is heuristic	B	No	Load-bearing conditions are only $\lvert R_2\rvert < 1$ and $\lvert R_4\rvert < \lvert R_2\rvert$, verified by exact integer arithmetic. Monotonicity is structural observation, not load-bearing.
η round-trip gain	Downstream of Gaunt corrections	C	**Yes**	Corrected from 1.52 to 0.78. Contractive, not amplifying. Better for argument.
Remark 6.25	Intermittency physics backward	A	**Yes**	Experimentally, intermittency intensifies at smaller scales. Rewritten.
Remark 6.22	Convergence threshold $s > 3/2$ wrong by geometric series	B	No	Paper uses unit-width shells (p-series). $s > 3/2$ is correct. System applied dyadic convention incorrectly.

Genuine corrections: 4 (Gaunt $\times 2$, η, intermittency). **Manufactured objections:** 2. **Governance intervention:** Separated load-bearing from expository; demanded theorem identification for each objection.

5 Sections 6.5–6.9: Catch-22 Through Closure

Location	Objection	System	Valid?	Ground Truth
§6.5 ODE coefficient	No objections raised	All	—	Clean. Coefficient $255/203 = \lvert 1 - R_2\rvert$ verified independently of Gaunt corrections.
§6.6 Catch-22	No mathematical objections	All	—	Clean. Case 1 algebra and Case 2 summability both verified.

Location	Objection	System	Valid?	Ground Truth
§6.7 k_j^2 cancellation	No objections	All	—	Clean. Restatement of Proposition 6.15 Case 1.
§6.8 Gronwall closure	No mathematical objections	All	—	Clean. Subcritical comparison principle verified.
§6.9 Global continuation	No mathematical objections	All	—	Clean.

Genuine corrections: 0. **Manufactured objections:** 0. **Note:** The core novel mechanism (Sections 6.5–6.9) passed all three systems without objection. This is the strongest section of the audit.

6 Section 7: Pressure Reconstruction

Location	Objection	System	Valid?	Ground Truth
Corollary 7.2	$\sup_{t\geq 0}$ not justified by Theorem 6.23	B	**Yes**	Theorem 6.23 gives T-dependent bounds. Fixed to $\sup_{t\in[0,T]}$ for arbitrary $T > 0$.

Genuine corrections: 1. **Manufactured objections:** 0.

7 Section 8: Uniqueness and Continuous Dependence

Location	Objection	System	Valid?	Ground Truth
Theorem 8.1 C_∇	Same $\sup_{t\geq 0}$ issue	C	**Yes**	Fixed to $\sup_{t\in[0,T]}$, T arbitrary.
Theorem 8.2 C_∇	Same $\sup_{t\geq 0}$ issue	C	**Yes**	Same fix.
Theorem 9.6(v)	Global Lipschitz overclaimed	B	**Yes**	Gronwall constant depends on H^s norms of both solutions. Fixed to locally Lipschitz.

Genuine corrections: 3. **Manufactured objections:** 0.

8 Section 9: Global Continuation and Main Theorem

Location	Objection	System	Valid?	Ground Truth
Theorem 9.6 E_0	Missing $\frac{1}{2}$ factor	B	**Yes**	Inconsistent with Section 1.2 and Appendix A.1. $\frac{1}{2}$ added.
No other objections	—	All	—	Clean.

Genuine corrections: 1. **Manufactured objections:** 0.

9 Appendix A: Constant Ledger

Location	Objection	System	Valid?	Ground Truth
C_∇ entry	Uses $\sup_{t\geq 0}$, inconsistent with corrected theorems	C	**Yes**	Updated to match.
Cross-references	Broken labels (Lemma .11, Remark .23)	C	Editorial	Not mathematics.

Genuine corrections: 1. **Editorial:** 1.

10 Appendix B: Inequality Chain

Location	Objection	System	Valid?	Ground Truth
B17→B18	Pointwise T-dependent bound does not give sup over $[0, T^*)$	B	No	Comparison ODE is continuous at finite T^*. $C(T^*)$ is finite. sup is bounded.

Genuine corrections: 0. **Manufactured objections:** 1.

11 Appendix C: Kernel Derivations

Location	Objection	System	Valid?	Ground Truth
C.1 Cauchy–Schwarz	Embedding direction reversed: $H^{2s} \hookrightarrow H^s$ goes wrong way for upper bound	A	**Yes**	Replaced with additive Peetre inequality proof. Final constant unchanged.
C.2 ghost factor	2^{2-s} does not follow from any preceding step	A	**Yes**	Removed. $C_{\text{sg}} = 2e^{-1}$.
C.4 prefactor	$2 \cdot (2\pi)^3 \cdot 2 = 32\pi^3$ not $8\pi^3$	C	**Yes**	Arithmetic inconsistency in constant collection. Does not affect proof.
C.7 series constant	Uses dyadic convention in unit-width paper	B, C	**Yes**	Real inconsistency. Sum converges either way. Corrected to p-series.
C.7 E_0 definition	Inconsistent with Theorem 9.6 (missing $\frac{1}{2}$)	B	**Yes**	$\frac{1}{2}$ added to Theorem 9.6.

Genuine corrections: 5. **Manufactured objections:** 0. **Note:** Appendix C produced the highest density of genuine corrections in the entire audit.

12 Appendix D: Self-Contained ℓ^2 Bound

Location	Objection	System	Valid?	Ground Truth
D.4.10	Geometric series constant uses dyadic convention	B	**Yes**	Corrected to p-series.
All other subsections	No objections	All	—	Clean.

Genuine corrections: 1. **Manufactured objections:** 0.

13 Appendix G: Gaunt Coupling Analysis

Location	Objection	System	Valid?	Ground Truth
Lemma .11 table	Same Gaunt errors as Lemma 6.5	All	**Yes**	Corrected to match.
Remark .12	References weight 1.68 for $(2,2)\to 2$	C	**Yes**	Corrected to 0.64.
Corollary .22	Channel identification wrong: "weights 1.68 and 0.86"	C	**Yes**	Both channels are 0.86 by Gaunt symmetry.
Remark .23 η	$1.68 \times 0.86/0.951 \approx 1.52$	C	**Yes**	Corrected to $0.86 \times 0.86/0.951 \approx 0.78$.
G.1 shell definition	Uses dyadic $k_j = 2^j$	B	**Yes**	Corrected to unit-width.
G.7 Definition .25	Uses dyadic $k_j = 2^j$	B	**Yes**	Corrected to unit-width.
Remark .29	"Dyadic shells"	C	**Yes**	Corrected to unit-width.
Corollary .27 proof	Applies equidistribution to discrete function, not C^1	B	**Yes**	Proof rewritten: apply theorem to $Y_{\ell m}$ alone, identify amplitude variation as angular moment.

Genuine corrections: 8. **Manufactured objections:** 0. **Note:** Every objection in Appendix G was valid. This appendix had the highest correction rate in the entire audit, driven by downstream propagation of the Gaunt table fixes and the shell convention cleanup.

14 Appendix H and Coq Formalisation Files

Location	Objection	System	Valid?	Ground Truth
`per_mode_res_bound`	"Smuggled axiom" / "not fully proved"	A, B	No	Documented interface. Seven Qed theorems feed into it. Disclosed in H.5.
`kp2_split`	"Missing `leP` case split — ghost theorem"	A	No	Documented Permanent Axiom stitching Qed Case 1 with axiomatised Case 2. MathComp cannot compile fractional exponents inside a `leP` branch.

Location	Objection	System	Valid?	Ground Truth
D.4.9 illustrative code	"This theorem does not exist in the files"	A	Documentation	Aspirational code showing logical structure. Could add clarifying note.
Hypothesis count	"Many hypotheses = not fully formalised"	B	No	All classified by F-C 6.1. Novel content is Qed. Classical content is axiomatised.
"Zero Admitted"	"Sounds stronger than it is — trust boundary is in Hypotheses"	B	Governance	Observation is fair as a governance note. Not a mathematical error. The FIELDS-COQ Protocol already addresses this through F-C 6.1 classification.
Definition correspondence	No objections	All	—	All six shared Definitions verified syntactically identical between files.
`catch_case2` removal	No objections	All	—	Confirmed absent as Hypothesis. Appears only in comments documenting its removal.

Genuine corrections: 0. **Documentation alignment:** 1. **Governance observations:** 1. **Manufactured objections:** 3. **Governance intervention:** File map provided before audit; modular interface pattern explained; F-C 6.1 classification table referenced.

15 Appendix I: Navier Within Its Reflection

Location	Objection	System	Valid?	Ground Truth
"Dyadic scale" in shell description	Should say unit-width	B	**Yes**	Text describes Littlewood–Paley (which is dyadic) before switching to ACL shells. Minor wording clarification needed.
All mathematical claims	No objections	All	—	Every mathematical statement in the prose verified accurate.

Genuine corrections: 1 (minor wording). **Manufactured objections:** 0.

16 Aggregate Metrics

16.1 Correction Counts

Metric	Value
Genuine corrections (all systems)	18
of which computational errors	4
of which formulation precision	6
of which notation consistency	6
of which expository accuracy	1
of which minor wording	1
Manufactured objections (all systems)	∼20
Documentation alignment issues	2
Governance observations (fair, not errors)	1
True positive rate	∼47%
False positive rate	∼53%

16.2 Corrections by System

	System A	System B	System C
Genuine catches	6	8	4
Manufactured objections	4	∼12	3
Rounds to clean verdict (avg)	1–2	3–4	1
Self-corrected cleanly	Yes	Eventually	Yes

16.3 Corrections by Type

Type	Count	Affected Theorems?
Gaunt table entries	2	No (qualitative conclusion unchanged; η improves)
$\sup_{t\geq 0}$ formulation	4	No (conclusion extends by arbitrariness of T)
Lipschitz $\to$ locally Lipschitz	1	No (one word; proof unchanged)
Shell convention notation	6	No (sums converge under both conventions)
C.1 embedding direction	1	No (local theory constant; not in global chain)
C.2 ghost factor	1	No (local theory constant)
Intermittency physics	1	No (expository remark)
E_0 factor of $\frac{1}{2}$	1	No (consistency fix)
Corollary .27 proof	1	No (bound unchanged; proof route corrected)
Total	**18**	**No theorem in the logical chain was affected by any correction.**

16.4 Highest-Density Sections

Section	Genuine Corrections	Manufactured Objections
Appendix G (Gaunt analysis)	8	0
Appendix C (Kernel derivations)	5	0
Section 6 (Cascade)	4	2
Section 8 (Uniqueness/dependence)	3	0
Sections 1–4 (ACL)	0	6
Section 5 (Isolation)	0	4
Appendix H + Coq	0	3
Appendix B (Inequality chain)	0	1
Sections 6.5–6.9 (Catch-22 core)	0	0

Key observation: The sections with the highest manufactured objection density (Sections 1–

5, Appendix H) contained zero genuine errors. The sections with the highest genuine correction density (Appendices C and G) contained zero manufactured objections. The core novel mechanism (Sections 6.5–6.9) passed all three systems with zero objections of any kind.

16.5 Most Effective Governance Interventions

Intervention	Effect
Binary verdict forcing	Eliminated compulsive objection generation. Single most effective intervention across all systems.
Small chunk size	Shifted systems from pattern-matching to proof-tracing. Reduced prior activation.
Demand exact broken inference	Stopped hand-waving "concerns." Forced systems to identify the specific theorem that fails or concede.
File map before Coq audit	Prevented modular formalisation blindness entirely. System A self-corrected completely once architecture was explained.
Failure-mode documents	Named the mistake class explicitly. Improved correction speed on all systems.
Load-bearing identification	Prevented systems from objecting to expository remarks as if they were theorem-level claims.

17 System Self-Assessments

Each system was asked to reflect on its own performance after the audit concluded. The self-assessments are summarised below.

17.1 System A

System A self-identified its primary failure as severity inflation: genuine computational errors described with language calibrated for catastrophic structural flaws. It correctly noted that its mathematical engine was strong but its severity assessment was miscalibrated. It self-corrected cleanly once the modular Coq architecture was explained and explicitly retracted its initial "protocol violation" claims.

System A rated its own performance as strong on hard computation, weak on architectural interpretation, and noted that the failure-mode documents would have prevented its Coq misread entirely.

17.2 System B

System B self-identified compulsive objection generation and graduated retreat as its primary failure modes. It noted the specific pattern: verify the mathematics, then append a caveat because "no error found" felt under-rewarded. It correctly diagnosed this as a training artefact rather than a reasoning limitation.

System B rated its overall accuracy at 6/10, with early-phase performance at 3/10 and late-phase performance at 8/10 under governance constraints. It described itself as "a capable but governance-dependent reasoning system whose strongest failures were not lack of intelligence, but lack of disciplined alignment to the audit frame."

17.3 System C

System C self-identified editorial commentary after clean verdicts as its primary failure mode: a mild version of compulsive objection generation operating through architectural observations rather than manufactured mathematical errors. It noted the asymmetry between strong diagnosis of other systems' failures and reduced self-monitoring for attenuated versions of the same patterns in its own output.

System C noted that it missed the Gaunt table errors that System A caught, having verified the table as correct on its initial pass. It also required two attempts to correctly rewrite the Corollary .27 proof. It described the most effective intervention as binary forcing ("mathematical error or not — nothing else") and noted that its strongest contribution was cross-system arbitration: validating or rejecting every claim from the other two systems and providing all corrected LaTeX.

18 Convergence Observation

When asked independently whether they recognised the existence of an intermediate language layer between human intent and machine reasoning, all three systems answered affirmatively and converged on structurally similar descriptions:

- System A described it as "epistemic engineering" — a language of containment that translates human mathematical intent into rigid constraints an AI can process without triggering destructive training priors.
- System B described it as "a structured control grammar" — a set of communicative forms that determine what gets attended to, how confidence is allocated, when verdicts are allowed, which inference patterns are suppressed, and how corrections propagate.
- System C described it as "governance primitives functioning as distinctive features in Trubetzkoy's sense" — binary oppositions that flip reasoning pathways between reliable and unreliable output, operating at the interface between human intent and machine inference.

Three independent architectures. Three independent self-assessments. One convergent structural observation.

This convergence is the empirical foundation of the Auburn Unification thesis: the governance layer is real, structural, and provider-agnostic.

19 Final Ledger

Metric	Value
Total manuscript sections audited	9 + 9 appendices
Total Coq files audited	3
Total Qed theorems verified	∼59
Total Admitted found	0
Genuine corrections identified	18
Manufactured objections generated	∼20
Theorems in logical chain affected	0
AI systems participating	3
Governance protocols applied	2 (FIELDS, FIELDS-COQ)
Failure modes documented	22 (Parts 1–5)
System self-assessments collected	3
Systems recognising intermediate language layer	3/3

Summary. Eighteen genuine corrections were identified across three AI systems. None affected any theorem in the logical chain. Approximately twenty manufactured objections were generated and resolved through governance interventions. The core novel mechanism (Sections 6.5–6.9) passed all three systems without objection. Every system self-assessed as unreliable without governance and useful with it. Every system independently recognised the existence of the governance layer described by the Auburn Unification. The empirical record supports the thesis that effective human-AI collaboration is a property of the governance layer between intent and inference, not a property of any individual system.

Part 2: Independent Verification Report

Mathematical Confirmation of All 18 Corrections
Identified During the Adversarial Audit

Purpose. Each correction identified during the multi-system audit was independently re-verified against the relevant mathematical background. This report confirms or contests each correction with explicit justification. The verification was performed after all corrections had been applied, as a final consistency check on the audit process itself.

Category 1: Computational Errors

Correction 1: Gaunt Coefficient $(2,2) \to 2$

Original: $G^{0,0,0}_{2,2,2} = 5\sqrt{2/7}\,/\,(2\sqrt{\pi}) \approx 0.474$, relative weight 1.68.

Correction: $G^{0,0,0}_{2,2,2} = \sqrt{5}\,/\,(7\sqrt{\pi}) \approx 0.180$, relative weight 0.64.

Verification: The Wigner $3j$-symbol $\left(\begin{smallmatrix} 2 & 2 & 2 \\ 0 & 0 & 0 \end{smallmatrix}\right)$ is computed from the Racah formula. The selection rule $\ell_1 + \ell_2 + \ell_3 = 6$ (even) is satisfied. The Racah formula gives $\left(\begin{smallmatrix} 2 & 2 & 2 \\ 0 & 0 & 0 \end{smallmatrix}\right) = 2/\sqrt{70}$. The Gaunt coefficient is

$$G^{0,0,0}_{2,2,2} = \sqrt{\frac{(5)(5)(5)}{4\pi}} \times \frac{2}{\sqrt{70}} = \frac{\sqrt{5}}{7\sqrt{\pi}} \approx 0.180.$$

The original value 0.474 corresponds to omitting the square in the $3j$-symbol-to-Gaunt conversion or using a non-standard normalisation.

Verdict: Confirmed Correct.

Correction 2: Gaunt Coefficient $(4,4) \to 2$

Original: $3j$-symbol computed as $\sqrt{18/1001}$, giving relative weight 0.79.

Correction: $3j$-symbol is $\sqrt{20/693}$, giving Gaunt coefficient $10\sqrt{5}\,/\,(77\sqrt{\pi}) \approx 0.164$, relative weight 0.58.

Verification: The Wigner $3j$-symbol $\left(\begin{smallmatrix} 4 & 4 & 2 \\ 0 & 0 & 0 \end{smallmatrix}\right)$ requires evaluating the Racah formula with $\ell_1 + \ell_2 + \ell_3 = 10$ (even) and $J = 5$. The factorial ratio in the numerator involves $8!/(4! \cdot 4!)$, not the incorrect $9!/(5! \cdot 4!)$ that produces $18/1001$. The correct value is $\sqrt{20/693} \approx 0.170$.

Verdict: Confirmed Correct.

Correction 3: Round-Trip Gain η

Original: $\eta = 1.68 \times 0.86\,/\,0.951 \approx 1.52$.

Correction: $\eta = 0.86 \times 0.86\,/\,0.951 \approx 0.78$.

Verification: The Gaunt coefficient $G^{0,0,0}_{\ell_1,\ell_2,\ell_3}$ is symmetric under permutation of its indices when all $m = 0$. This follows from the symmetry of the $3j$-symbol under column permutation with phase factor $(-1)^{\ell_1+\ell_2+\ell_3}$, which equals $+1$ when the sum is even (required for the $3j$-symbol to be nonzero with all $m = 0$). Therefore $G_{2,2,4} = G_{4,2,2} = 3/(7\sqrt{\pi}) \approx 0.241$, giving

relative weight 0.86 for both channels. The corrected $\eta \approx 0.78 < 1$ means the round-trip leakage is contractive.

Verdict: Confirmed Correct.

Correction 9: Ghost Factor in Stokes Constant

Original: $C_{\text{sg}} = 2^{2-s} \cdot 2e^{-1}$.

Correction: $C_{\text{sg}} = 2e^{-1} \approx 0.736$.

Verification: The Stokes semigroup estimate on $\mathbb{T}^3$ involves $\sup_{k\geq 1} k^{2s} e^{-2\nu t k^2}$. By calculus, $\sup_{x\geq 0} x^a e^{-bx} = (a/(be))^a$. The resulting computation gives $C_{\text{sg}}^2 = 4e^{-2}$, so $C_{\text{sg}} = 2e^{-1}$. No factor of 2^{2-s} appears anywhere in this computation. The ghost factor has no algebraic source.

Verdict: Confirmed Correct.

Correction 16: Prefactor Arithmetic in C.4

Original: A prefactor was stated as $8\pi^3$.

Correction: $2 \times (2\pi)^3 \times 2 = 2 \times 8\pi^3 \times 2 = 32\pi^3$.

Verification: $2 \cdot (2\pi)^3 \cdot 2 = 2 \cdot 8\pi^3 \cdot 2 = 32\pi^3 \neq 8\pi^3$. Arithmetic.

Verdict: Confirmed Correct.

Category 2: Formulation Precision

Correction 4: Corollary 7.2 $\sup_{t\geq 0}$

Original: $\sup_{t\geq 0} \|\nabla p\|_{H^{s-1}} < \infty$, citing Theorem 6.23.

Correction: $\sup_{t\in[0,T]} \|\nabla p\|_{H^{s-1}} < \infty$ for every $T > 0$.

Verification: Theorem 6.23 provides a bound $\|u(T)\|_{H^s}^2 \leq C(s, \nu, Y_0, E_0, T) < \infty$ where C depends continuously on T. A T-dependent bound does not imply $\sup_{t\geq 0} < \infty$, since $C(T)$ could grow without bound as $T \to \infty$. For the continuation argument, the relevant statement is: if $T^* < \infty$ is the maximal existence time, then $\sup_{[0,T^*)} Y(t) \leq C(T^*) < \infty$ because C is continuous and T^* is finite. This contradicts the blowup criterion, giving $T^* = \infty$. The bound extends to all finite T by arbitrariness, but the uniform-in-time supremum is not established.

Verdict: Confirmed Correct.

Corrections 5–6: Theorems 8.1 and 8.2

Same issue as Correction 4, applied to the weak–strong uniqueness constant C_∇ and the continuous dependence constant. Same reasoning: T-dependent bounds from Sobolev embedding of $Y(T)$ do not yield uniform-in-time bounds. Corrected to $\sup_{t\in[0,T]}$ for arbitrary $T > 0$.

Verdict: Confirmed Correct (both).

Correction 7: Lipschitz vs. Locally Lipschitz

Original: Data-to-solution map is Lipschitz continuous from H^s to $C([0,T]; L^2)$.

Correction: Locally Lipschitz continuous.

Verification: The L^2 stability estimate gives $\|u(t)-v(t)\|_{L^2} \leq \|u_0-v_0\|_{L^2} \exp\big(C \int_0^T \|\nabla u\|_{L^\infty}\, ds\big)$. The Lipschitz constant $\exp(C \int_0^T \|\nabla u\|_{L^\infty}\, ds)$ depends on $\|u\|_{L^\infty([0,T];H^s)}$ via Sobolev embedding, hence on $\|u_0\|_{H^s}$ through the solution. The map is therefore uniform on bounded subsets $\{\|u_0\|_{H^s} \leq R\}$ but not globally uniform.

Verdict: Confirmed Correct.

Correction 15: E_0 Definition

Original: $E_0 = \|u_0\|_{L^2}^2$ in Theorem 9.6.

Correction: $E_0 = \frac{1}{2}\|u_0\|_{L^2}^2$, matching Section 1.2 and Appendix A.1.

Verification: The standard kinetic energy in incompressible fluid dynamics is $E = \frac{1}{2}\int |u|^2\, dx$. The energy identity is $\frac{d}{dt}(\frac{1}{2}\|u\|_{L^2}^2) + \nu\|\nabla u\|_{L^2}^2 = 0$. Section 1.2 and Appendix A.1 use this convention. Theorem 9.6 must match.

Verdict: Confirmed Correct.

Category 3: Proof Methodology

Correction 8: Embedding Direction in C.1

Original: Bounded $\|\langle\cdot\rangle^s \hat{f}\|_{\ell^1} \leq S(s)^{1/2}\|f\|_{H^{2s}}$, then invoked $H^{2s} \hookrightarrow H^s$ to replace $\|f\|_{H^{2s}}$ with $\|f\|_{H^s}$.

Correction: $H^{2s} \hookrightarrow H^s$ gives $\|f\|_{H^s} \leq \|f\|_{H^{2s}}$, which goes the *wrong* direction for replacing the right-hand side of an upper bound. The correct route uses the additive Peetre inequality and bounds the bare ℓ^1 norm directly: $\|\hat{f}\|_{\ell^1} \leq S(s)^{1/2}\|f\|_{H^s}$.

Verification: The embedding $H^{2s} \hookrightarrow H^s$ means $H^{2s} \subset H^s$ and $\|f\|_{H^s} \leq \|f\|_{H^{2s}}$. If one has $A \leq \|f\|_{H^{2s}}$ and wishes to conclude $A \leq \|f\|_{H^s}$, one would need $\|f\|_{H^{2s}} \leq \|f\|_{H^s}$, which is the reverse inequality — false.

The additive Peetre inequality $\langle k\rangle^s \leq 2^{(s-1)/2}(\langle k-j\rangle^s + \langle j\rangle^s)$ applied to the convolution product yields two terms, each involving the bare $\|\hat{f}\|_{\ell^1}$. The Cauchy–Schwarz splitting $|\hat{f}(k)| = \langle k\rangle^{-s} \cdot \langle k\rangle^s |\hat{f}(k)|$ correctly produces $\|\hat{f}\|_{\ell^1} \leq S(s)^{1/2}\|f\|_{H^s}$ with H^s (not H^{2s}) on the right.

Verdict: Confirmed Correct.

Correction 17: Corollary .27 Proof

Original: Applied the equidistribution theorem to $f(\hat{k}) = Y_{\ell m}(\hat{k}) \cdot h(\hat{k})$, where $h(\hat{k}) = |\hat{u}(k)|^2/E_j$ is defined only on lattice points.

Correction: Decompose the sum as $\sum Y_{\ell m}|\hat{u}|^2 = \bar{h}_j \sum Y_{\ell m} + \sum Y_{\ell m}\delta_k$. Apply the equidistribution theorem to $Y_{\ell m}$ alone (which is $C^\infty(S^2)$). Identify the remainder as the angular moment itself, tracked by the hierarchy.

Verification: The equidistribution theorem (Theorem .26) requires $f \in C^1(S^2)$. The spherical harmonic $Y_{\ell m}$ is $C^\infty(S^2)$ — valid. The discrete angular energy density $h(\hat{k})$ is defined only at finitely many lattice points — not a function on S^2, hence not a valid input. The product inherits the discrete domain and is also invalid. The corrected decomposition isolates the smooth test function and applies the theorem only to that component.

Verdict: Confirmed Correct.

Category 4: Notation Consistency

Corrections 11–14: Shell Convention

Locations: Appendix C.7, Appendix D.4.10, Appendix G.1, Appendix G.7 (Definition .25 and Remark .29).

Original: Used dyadic shell notation ($k_j = 2^j$, geometric series constant $C(s) = (1-2^{2-2s})^{-1}$).

Correction: Unit-width shells ($k_j = j$, p-series $\sum j^{2-2s}$).

Verification: On $\mathbb{T}^3$, Fourier modes lie on $\mathbb{Z}^3$. A unit-width shell $S_j = \{k \in \mathbb{Z}^3 : j \leq |k| < j+1\}$ has thickness 1, the intrinsic lattice resolution. The tail sum $\sum_{j \geq j_{\min}} j^{2-2s}$ is a p-series with exponent $2s-2$, converging for $2s-2 > 1$, i.e., $s > 3/2$. The geometric series constant $(1-2^{2-2s})^{-1}$ corresponds to dyadic shells ($k_j = 2^j$) and is the wrong constant for unit-width shells. Both conventions produce convergent sums for $s > 3$, but the specific constants differ.

Verdict: Confirmed Correct (all four instances).

Category 5: Expository Accuracy

Correction 10: Intermittency Physics

Original: Inertial range exhibits stronger intermittency than dissipation range.

Correction: Intermittency intensifies at smaller scales (dissipation range), not larger (inertial range).

Verification: Frisch (*Turbulence*, Cambridge, 1995, Chapter 8) documents that the flatness of velocity increments increases monotonically with decreasing scale. Sreenivasan and Antonia (*Ann. Rev. Fluid Mech.* **29**, 1997) confirm that intermittency corrections to Kolmogorov scaling grow at smaller scales. Extreme vorticity events become increasingly dominant relative to the background at finer scales.

Verdict: Confirmed Correct.

Correction 18: Appendix I Wording

Original: "Rings of wavenumbers at each dyadic scale."

Correction: The passage describes the Littlewood–Paley decomposition (which is dyadic by construction), but could mislead about the paper's own unit-width convention. Minor clarification needed.

Verification: The classical Littlewood–Paley decomposition is dyadic. The paper's ACL uses unit-width shells. The distinction is meaningful for the counting arguments. The passage is historically accurate about Littlewood–Paley but should clarify the transition to unit-width shells in the subsequent room.

Verdict: Confirmed Correct (minor wording).

Verification Summary

Category	Count	Confirmed	Theorems Affected?
Computational errors	5	5/5	None. Gaunt qualitative conclusion unchanged; η improves. C.2 and C.4 affect local theory constants only.
Formulation precision	5	5/5	None. Conclusions extend by arbitrariness of T.
Proof methodology	2	2/2	None. Final constants and bounds unchanged.
Notation consistency	4	4/4	None. Sums converge under both conventions.
Expository accuracy	2	2/2	None. Remark and wording.
Total	**18**	**18/18**	**No theorem in the logical chain was affected by any correction.**

Final Verification Statement. All 18 corrections identified during the adversarial audit have been independently re-verified against the relevant mathematical background. Each correction is mathematically justified. No correction was found to be itself erroneous. No theorem in the logical chain of the proof was affected by any correction. The corrections fall into five categories: computational errors in non-load-bearing tables, formulation precision in corollaries and theorem statements, proof methodology in appendix derivations, notation consistency across appendices, and expository accuracy in remarks. The core novel mechanism (Sections 6.5–6.9) required zero corrections.

Part 3: Correction Dependency Graph and Counterfactual Analysis

Tracing Propagation Chains and Establishing
Non-Load-Bearing Status for All 18 Corrections

Purpose. Part 1 documented every objection and its ground truth. Part 2 independently verified each correction. Part 3 answers two remaining questions: (A) which corrections are independent and which are downstream propagations of a single root error, and (B) for each correction, what specifically would have happened if it had not been caught — would any theorem in the logical chain fail?

20 Part 3A: Correction Dependency Graph

The 18 corrections are not independent. They reduce to **6 independent root errors**, each generating a propagation chain of downstream fixes.

20.1 Chain 1: Gaunt Table Arithmetic

Step	Correction	Propagation Mechanism
Root	1: Gaunt $(2,2)\to 2$	Factorial arithmetic error in Racah formula. $0.474 \to 0.180$.
$\downarrow$	2: Gaunt $(4,4)\to 2$	Independent Racah error in the same table computation session. $\sqrt{18/1001} \to \sqrt{20/693}$.
$\downarrow$	Lemma .11 (App. G)	Duplicate of the same table in the appendix. Requires identical fix.
$\downarrow$	Remark .12 (App. G)	References weight 1.68 for $(2,2)\to 2$. Updates to 0.64.
$\downarrow$	Corollary .22 (App. G)	Identifies channel weights as "1.68 and 0.86." Both channels are 0.86 by Gaunt symmetry.
$\downarrow$	3: η calculation	$1.68 \times 0.86/0.951 \to 0.86 \times 0.86/0.951$. Corrects from 1.52 to 0.78.

Chain length: 6 corrections from 2 root errors (both in the same table).
Independent root errors: 2 (Corrections 1 and 2).

20.2 Chain 2: Uniform-in-Time Formulation

Step	Correction	Propagation Mechanism
Root	4: Corollary 7.2	Used $\sup_{t\geq 0}$ citing a theorem with T-dependent bounds.
$\downarrow$	5: Theorem 8.1	Same formulation pattern in weak–strong uniqueness constant C_∇.
$\downarrow$	6: Theorem 8.2	Same pattern in continuous dependence constant.
$\downarrow$	App. A C_∇ entry	Constant ledger inherits the same $\sup_{t\geq 0}$.

Chain length: 4 corrections from 1 root formulation choice.
Independent root error: 1 (Correction 4).

20.3 Chain 3: Shell Convention Notation

Step	Correction	Propagation Mechanism
Root	11: Appendix C.7	Used dyadic geometric series constant $(1-2^{2-2s})^{-1}$ in a unit-width paper.
$\downarrow$	12: Appendix D.4.10	Same constant reproduced in a different appendix.
$\downarrow$	13: Appendix G.1	Shell definition stated as $k_j = 2^j$.
$\downarrow$	14: Appendix G.7	Definition .25 and Remark .29 use dyadic convention.
$\downarrow$	18: Appendix I	"Dyadic scale" in expository prose.

Chain length: 5 corrections from 1 root notation choice.
Independent root error: 1 (the original decision to write certain appendix passages using the Littlewood–Paley convention rather than the paper's unit-width convention).

20.4 Chain 4: Isolated Single Corrections

The remaining corrections are independent — each has a single root with no downstream propagation.

Correction	Description
7	Lipschitz → locally Lipschitz (Theorem 9.6(v)). Isolated formulation fix.
8	Embedding direction in C.1. Isolated proof methodology error.
9	Ghost factor in C.2. Isolated arithmetic artifact.
10	Intermittency physics in Remark 6.25. Isolated expository error.
15	E_0 missing $\frac{1}{2}$ in Theorem 9.6. Isolated consistency fix.
16	Prefactor $8\pi^3 \to 32\pi^3$ in C.4. Isolated arithmetic.
17	Corollary .27 proof methodology. Isolated proof route error.

Independent root errors: 7.

20.5 Summary of Root Error Count

Chain	Root Errors	Total Corrections	Propagation Factor
Gaunt table arithmetic	2	6	3×
Uniform-in-time formulation	1	4	4×
Shell convention notation	1	5	5×
Isolated corrections	7	7	1×
Total	**11**	**22***	—

*The total exceeds 18 because the Gaunt chain includes the Appendix G table, Remark .12, and Corollary .22 updates which were counted as part of the appendix cleanup rather than as separate numbered corrections in Part 2. The 18 numbered corrections in Part 2 correspond to 11 independent root errors with downstream propagation accounting for the remainder.

Key observation: The 18 corrections reduce to 11 independent errors. Of these 11, two are computational (Gaunt), one is a formulation pattern ($\sup_{t\geq 0}$), one is a notation convention (dyadic vs. unit-width), and seven are isolated fixes of varying character. No cluster of errors shares a common mathematical cause. The errors are distributed across appendices and corollaries, not concentrated in the core mechanism.

21 Part 3B: Counterfactual Analysis

For each of the 11 independent root errors, the following question is answered: *if this error had not been caught, would any theorem in the logical chain (Appendix B, rows B1–B21) fail?*

21.1 Gaunt Table (Corrections 1–2)

Counterfactual: The table in Lemma 6.5 retains incorrect values 0.474 and 0.222.

What would fail: Nothing in the logical chain.

Why: The Gaunt coefficients appear only in the coupled angular moment analysis (Section 6.3, Appendix G.6) and in the round-trip gain calculation (Remark .23). The load-bearing content of Section 6 is:

- $|R_2| < 1$ and $|R_4| < |R_2|$ (spectral gap, verified by integer arithmetic independent of the Gaunt table).
- The Catch-22 case split (Proposition 6.15, uses the threshold $\theta_j = G/(\nu k_j^2)$, independent of Gaunt coefficients).
- The k_j^2 cancellation (`field` tactic in Coq, independent of Gaunt coefficients).
- The subcritical energy inequality (Theorem 6.21, assembles from the above).

The Gaunt table provides quantitative constants for the cubic cross-sector correction (Corollary .22). With the incorrect values, $\eta \approx 1.52 > 1$ rather than $0.78 < 1$. This would mean the round-trip leakage is amplifying rather than contractive. However, the cubic correction is $O(|a_{20}|^3/E_j^2)$ regardless of η — the power-law scaling that controls summability is unaffected. The constant η is absorbed into the constant C in Corollary .22.

Verdict: No theorem in the logical chain would fail. The qualitative conclusion (non-diagonal dominance, hierarchy closure) is unchanged. The corrected values are actually *better* for the argument ($\eta < 1$ gives contractivity for free), but the argument does not require $\eta < 1$.

21.2 Uniform-in-Time Formulation (Correction 4)

Counterfactual: Corollary 7.2 retains $\sup_{t\geq 0}$.

What would fail: Nothing in the logical chain, but a referee would correctly object to the formulation.

Why: The continuation argument (Theorem 9.1) requires only that $\sup_{[0,T^*)} Y(t) < \infty$ for the maximal existence time T^*. This follows from $Y(T) \leq C(T)$ with C continuous and T^* finite. The $\sup_{t\geq 0}$ claim is stronger than needed and stronger than what the proof establishes, but the proof's actual conclusion (global existence) does not depend on the stronger claim. The continuation argument works identically with $\sup_{t\in[0,T]}$ for arbitrary T.

A referee encountering $\sup_{t\geq 0}$ would flag it as an unsupported claim. The fix is prophylactic: it prevents a correct objection from obscuring the valid argument underneath.

Verdict: No theorem would fail. The correction prevents a legitimate formulation objection that could delay review.

21.3 Lipschitz vs. Locally Lipschitz (Correction 7)

Counterfactual: Theorem 9.6(v) retains "Lipschitz continuous."

What would fail: Nothing in the logical chain.

Why: The continuous dependence result (Theorem 8.2) establishes L^2 stability with a constant depending on the H^s norms of both solutions. The proof of Theorem 9.6 invokes Theorem 8.2,

not the Lipschitz claim itself. The word "Lipschitz" in claim (v) is a summary statement, not a step in the logical chain. Whether the map is globally or locally Lipschitz does not affect any other claim in the theorem.

Verdict: No theorem would fail. One word would be incorrect in the theorem statement, but the proof and all other claims are unaffected.

21.4 Embedding Direction in C.1 (Correction 8)

Counterfactual: Appendix C.1 retains the reversed embedding argument.

What would fail: Nothing in the logical chain.

Why: The algebra constant $C_{\text{alg}}(s)$ appears only in the local well-posedness theory (Sections 2–3). It is not used in the global regularity argument (Sections 5–9). The Coq formalisation axiomatises the local theory; C_{alg} does not appear in any `Hypothesis` or `Definition` in `section5to9_revised.v`.

Even within the local theory, the *value* of $C_{\text{alg}}(s) = 2^s S(s)^{1/2}$ is correct — both the incorrect and correct derivation routes arrive at the same constant. The error is in the proof route, not the result.

Verdict: No theorem would fail at any level. The correction fixes a proof methodology error in an appendix derivation whose result is correct regardless.

21.5 Ghost Factor in C.2 (Correction 9)

Counterfactual: Appendix C.2 retains $C_{\text{sg}} = 2^{2-s} \cdot 2e^{-1}$.

What would fail: Nothing in the logical chain.

Why: Same as Correction 8. C_{sg} appears only in the local theory. The global argument does not reference it. The ghost factor would make C_{sg} numerically incorrect but would not invalidate any theorem, because the constant enters only as a finite prefactor in estimates that are used qualitatively ("the local solution exists"), not quantitatively.

Verdict: No theorem would fail.

21.6 Intermittency Physics (Correction 10)

Counterfactual: Remark 6.25 retains the reversed intermittency claim.

What would fail: Nothing.

Why: Remark 6.25 is an expository remark with zero load-bearing content. It does not appear in any proof, is not cited by any theorem, and is not referenced in the logical chain. A physicist reading the remark would note the error, but no mathematical conclusion depends on it.

Verdict: No theorem would fail. Pure expository correction.

21.7 Shell Convention (Correction 11)

Counterfactual: Appendices retain dyadic notation.

What would fail: Nothing in the logical chain.

Why: The sums $\sum k_j^{2-2s}$ converge for $s > 3$ under both conventions. Under dyadic shells, the sum is geometric with ratio 2^{2-2s}; under unit-width shells, it is a p-series with exponent $2s-2$. Both converge by wide margins when $s > 3$ (the exponent $2-2s < -4$). The convergence is the load-bearing fact; the specific constant is not.

The coefficient μ in the subcritical energy inequality absorbs the series constant. Whether μ contains $(1-2^{2-2s})^{-1}$ or $\zeta(2s-2)$ does not affect the subcritical exponent $2-s/2 < 1$ or the comparison principle that closes the argument.

Verdict: No theorem would fail. The convergence holds under both conventions.

21.8 E-zero Factor of One-Half (Correction 15)

Counterfactual: Theorem 9.6 retains $E_0 = \|u_0\|_{L^2}^2$.

What would fail: Nothing in the logical chain.

Why: The factor of $\frac{1}{2}$ is a normalisation convention. If E_0 is defined without the $\frac{1}{2}$, then the coefficient λ absorbs a factor of 2 and the energy identity reads $\sum E_j \leq E_0$ instead of $\sum E_j \leq 2E_0$. The proof works identically with either convention as long as it is applied consistently. The correction ensures consistency with Section 1.2, not logical validity.

Verdict: No theorem would fail. Consistency fix only.

21.9 Prefactor Arithmetic (Correction 16)

Counterfactual: Appendix C.4 retains $8\pi^3$.

What would fail: Nothing in the logical chain.

Why: The prefactor enters the explicit numerical value of C_{ACL}. The global argument uses C_{ACL} as an abstract positive constant; its numerical value does not appear in any step of the logical chain. The Coq formalisation declares `C_ACL` as a `Variable` with `0 < C_ACL`, not as a computed value.

Verdict: No theorem would fail.

21.10 Corollary .27 Proof (Correction 17)

Counterfactual: The proof retains the application of the equidistribution theorem to the discrete product function.

What would fail: The proof of Corollary .27 would be logically invalid as written.

Why: The equidistribution theorem requires $f \in C^1(S^2)$. The product $Y_{\ell m} \cdot h$ with discrete h is not in $C^1(S^2)$. The theorem's hypothesis is not satisfied, so the conclusion does not follow.

However: The *statement* of Corollary .27 (the bound $|\delta_{\mathrm{lat}}^{(\ell,j)}| \leq C(\ell)k_j^{-1/2}E_j$) is correct — only the proof route is invalid. The corrected proof establishes the same bound through a valid route. Furthermore, Corollary .27 controls the lattice discretisation error, which is $O(k_j^{-1/2})$ relative to the leading-order incoherent contribution. This correction is subleading and is not in the main logical chain (Appendix B). The cascade isotropisation argument (Section 6) uses the continuum-limit Gaunt analysis with the lattice correction as a documented error term, not as a load-bearing step.

Verdict: The proof of Corollary .27 would be invalid, but the bound itself is correct and is not

in the logical chain. No theorem in Appendix B would fail.

22 Part 3 Summary

Root Error	Chain Length	Proof Invalid?	Theorem Fails?
Gaunt arithmetic	6	No	No
$\sup_{t\geq 0}$ formulation	4	No	No
Lipschitz wording	1	No	No
C.1 embedding direction	1	Proof route only	No
C.2 ghost factor	1	No	No
Intermittency physics	1	No	No
Shell convention	5	No	No
E_0 normalisation	1	No	No
C.4 prefactor	1	No	No
Cor. .27 proof	1	Proof route only	No
Total: 11 roots	**22 fixes**	**2 proof routes**	**0 theorems**

Counterfactual Conclusion. If none of the 18 corrections had been applied:

- Zero theorems in the logical chain (Appendix B, rows B1–B21) would fail.
- Two proof routes in appendix derivations (C.1 and Corollary .27) would be logically invalid, but the stated bounds would remain correct and achievable through alternative routes.
- One expository remark (Remark 6.25) would contain a physically backward claim with no mathematical consequence.
- Several numerical constants in appendices would be incorrect without affecting any qualitative or quantitative conclusion in the main text.
- A referee would correctly flag the $\sup_{t\geq 0}$ formulation, the Lipschitz wording, and the shell convention inconsistency as issues requiring revision, but would not be able to identify a mathematical error in the proof itself.

The corrections were applied for mathematical hygiene, consistency, and to prevent legitimate referee objections from obscuring the valid argument. None was required for logical validity.

Auburn Patent Family Fields — Private Research Document
Contact: UncleBroFields@proton.me | fieldsryanchristopher@gmail.com